RiskPath Demo Runbook

Live walkthrough script for the executive demo

Frontend: riskpath-clinician-companion.lovable.app · Core demo ~5 minutes · Format: DO / SEE / SAY

---

5 Minutes Before You Present — Pre-Flight

Do this right before you walk in. It guarantees a smooth, lag-free demo.

1. Open **https://riskpath-clinician-companion.lovable.app** on your iPad.
2. Confirm the green **"Online"** badge in the top-right corner. If it shows red, reload the page once.
3. Tap **Patient #1**, then scroll the cards and tap **Patient #5**. This warms up the model so there is no lag during the real demo.
4. Tap **Re-explain** once. This warms up the SHAP explanation engine.
5. Reload the page so you start clean on Patient #1.
6. Leave it on the **Predict** screen, ready to go.

Your Patient Roster (memorize these)

The five sample patients are ordered left-to-right from lowest to highest risk. No data entry needed — you just tap a card.

Card	Diagnosis	Discharge to	Risk	Band
#1	Symptoms	Home	0.020	LOW
#2	Respiratory	Home Health Care	0.100	LOW
#3	Circulatory	Rehab	0.380	MODERATE
#4	(chapter A)	Skilled Nursing	0.680	HIGH
#5	(chapter Z)	Skilled Nursing	0.898	VERY HIGH

Patient #1 is your low-risk baseline. Patient #5 is your dramatic high-risk case (12 prior admissions in 6 months). Those two are the stars.

Step 1 — Frame the Problem (30 seconds)

DO: Before touching the screen, set the stage.

SAY:

> *Hospital readmissions cost the US health system about 26 billion dollars a year, and CMS penalizes us for them through the Readmissions Reduction Program. The problem isn't that we don't care — it's that on a busy discharge day, we can't tell which patients are actually at risk until it's too late. RiskPath solves that. Let me show you.*

Step 2 — Establish It's Real (15 seconds)

DO: Point to the green **"Online"** badge in the top-right.

SAY:

> *Everything you're about to see is a live model running in production — not a slideshow, not fake data. Every number is a real prediction.*

Step 3 — The Low-Risk Baseline (45 seconds)

DO: Tap **Patient #1** (Symptoms, going Home).

SEE: The gauge settles at **0.020** with a green **LOW** pill.

SAY:

> *Here's a patient discharged home after a symptom-related admission. RiskPath puts their 30-day readmission risk at 2 percent. This is a green-light patient — routine discharge planning, standard follow-up. We don't need to spend scarce care-management resources here.*

Step 4 — The Dramatic Contrast (1 minute) — YOUR MONEY MOMENT

DO: Scroll the patient cards to the right, then tap **Patient #5** (Skilled Nursing).

SEE: The gauge sweeps up to **0.898**, the arc fills green-yellow-orange-red, and a red **VERY HIGH** pill appears.

SAY:

> *Now watch the same model on a different patient. Ninety percent risk. This patient is very likely to be back in our hospital within 30 days. If they walk out the door with a standard discharge packet, we've almost certainly missed our chance to intervene.*

DO: Pause. Let the red gauge sit on screen for a beat — it's visually striking.

Step 5 — Explain WHY (1.5 minutes) — YOUR STRONGEST DIFFERENTIATOR

DO: Tap **"Re-explain"** on the Top 5 Risk Drivers card. Wait about 3 seconds for the bars to appear.

SEE: Five red bars — Prior Readmission Count, Admission Frequency times Recency, Prior Admissions over 6 months, and related utilization features.

SAY:

> *This is what makes RiskPath different from a black box. It doesn't just say "high risk" — it tells us exactly why. Look at the top drivers: this patient has been admitted twelve times in the last six months. The model is seeing a clear frequent-flyer pattern. That's not a mystery score — it's an actionable signal that points our care team straight at an intensive transitions-of-care program.*

SAY (the trust line):

> *And when a physician asks "why should I trust this?", I can show them. Every prediction is explainable at the patient level, in clinical terms they recognize.*

Step 6 — Turn Risk Into Action (45 seconds)

DO: Scroll down to the **Recommended Care Pathway** card.

SEE: Band-specific recommendations (phone outreach within 48 to 72 hours, transportation plan, patient-portal enrollment) with a footer citing Project RED and IHI BOOST.

SAY:

> *And it closes the loop. For each risk level, RiskPath maps to evidence-based interventions — sourced from Project RED and IHI's BOOST guidelines, and fully customizable to our own protocols. Risk score in, care plan out.*

Step 7 — Trust and Governance (1 minute) — WHAT WINS THE EXECUTIVE

DO: Tap **Model Card** in the left sidebar.

SEE: Model name xgboost-v7-seed0, AUROC 0.7929, 50 features, and the Methodology and Intended Use sections.

SAY:

> *For your governance committee: this is a peer-reviewed model with a validated AUROC of 0.79, trained on 244,000 admissions. And critically —*

DO: Scroll to the **Intended Use and Limitations** section.

SAY:

> *— it's transparent about its own limits. This is decision support, not autonomous decision-making. It explicitly states what it should and shouldn't be used for, and that any institution needs to validate it locally before relying on it. That responsible framing is the difference between a tool your medical staff will adopt and one your general counsel will block.*

Step 8 — Close (30 seconds)

DO: Tap back to **Predict**.

SAY:

> *Three takeaways. One — this is a real, validated model, not a demo. Two — every prediction is explainable in terms clinicians can act on. Three — it's already running in production, and we could stand up a pilot on our own data in a matter of weeks. The question isn't whether we can predict readmission risk. It's whether we want to start acting on it before patients walk out the door.*

DO: Stop talking. Let them respond.

Optional Extensions (only if engaged and time allows)

Show the model discriminates smoothly: Tap #1, then #3, then #5 in sequence. Say: "It's not just high or low — it grades risk across the whole spectrum."

Show interactivity: Scroll to the Patient Features panel, edit a value (for example, bump Prior Admissions), and watch the risk re-score live. Say: "It responds to the clinical reality in real time." Only do this if you've rehearsed it.

Show batch scoring: Click Batch Score and upload a CSV. Say: "And it scales — score your entire discharge census in seconds."

Likely Questions and Your Answers

Is patient data stored?

No — the backend is stateless. Predictions flow through and nothing is persisted. No PHI in logs, no database.

What was it trained on?

MIMIC-IV — a public, de-identified critical-care dataset of 244,000 admissions from Beth Israel Deaconess. For a real pilot, we'd retrain on our own population.

How accurate is it?

AUROC of 0.79 — meaningfully better than chance, in line with published readmission models. And it's slightly under-confident at the high end, which is the safe direction clinically.

Can we use this tomorrow?

Not as-is on real patients — it needs local validation and an IRB conversation first. But the platform is built and deployed. The gap is validation, not engineering.

What does it cost to run?

Effectively nothing right now — under 5 dollars a month of cloud infrastructure. The cost is in validation and integration, not the technology.

Technical and Infrastructure Questions

These come from a CIO, CTO, or IT director in the room. Answer plainly and confidently.

Where is the backend deployed?

The backend runs on Railway — a managed cloud platform. It's a Python FastAPI service packaged as a Docker container, and it auto-deploys from GitHub every time we push a change. The model itself — an XGBoost classifier with SHAP explanations — is wrapped inside that service.

Where is the frontend deployed?

The frontend is a React application hosted on Lovable, which deploys to Cloudflare's global edge network. That's why it loads quickly from anywhere and runs on any device — including this iPad — with nothing to install.

What's the technology stack?

Backend is Python — FastAPI, XGBoost, and SHAP. Frontend is React with the TanStack framework. Everything is in version control on GitHub and deploys automatically on every change. It's a standard, modern, maintainable stack — nothing exotic that would lock us in or be hard to staff.

How does it scale?

A single prediction takes about 80 milliseconds, and a SHAP explanation about 200. It scales horizontally — we add instances as load grows. For a hospital-scale pilot, the current setup handles a full discharge census comfortably, and we can batch-score a hundred patients in about a second and a half.

Is it HIPAA-compliant and secure?

The current deployment is a research demo on de-identified public data, so HIPAA doesn't apply yet. For a real deployment with PHI, we'd move to a HIPAA-eligible environment under a Business Associate Agreement — Railway, AWS, Azure, and Google Cloud all offer that. The application doesn't change; only the hosting tier does.

What happens to patient data?

Nothing is stored. The backend is stateless — a prediction flows through and the data is discarded immediately. There's no database and no logging of patient features. That's a deliberate design choice that simplifies the privacy and security story.

Can it integrate with our EHR — Epic or Cerner?

Yes. At its core it's an API. Any system that can make a secure web call can send a patient record and get back a risk score and explanation. Integration through FHIR or an interface engine is a well-trodden path — it's an integration project, not a rebuild.

How long did this take to build?

The platform — backend, frontend, and deployment — came together in days, on top of a pre-existing, peer-reviewed research model. That's the key point: the engineering is largely done. The remaining work for a real deployment is clinical validation and EHR integration, not building from scratch.

What if it goes down — who maintains it?

Both services auto-restart on failure and auto-redeploy from GitHub, with health checks built in. For a production pilot we'd layer on monitoring and on-call coverage, but the foundation — version control, containerized deployments, automated health checks — is already in place.

If Something Breaks Mid-Demo

Blank prediction or "Offline" badge: Say "Let me reload that," reload the page, and tap the patient again. You pre-warmed it, so this is unlikely.

Slow first prediction: Say "It's spinning up," wait 3 seconds, and it'll resolve.

Total failure: You have the platform-guide PDF as backup — walk through it verbally. Nobody will know the difference. Confidence carries the room.